# Support Vector Machines

- You have studied some well-known classification models, i.e. logistic regression and naive Bayes. Though these models are widely used in industries such as banking, e-commerce etc. and applications such as default prediction and text classification, they find limited use in more complex classification problems, such as image classification.
- SVMs, as you will see, are capable of dealing with quite complex problems, where models such as logistic regression typically fail. SVMs have been extensively used for solving complex classification problems such as image recognition, voice detection etc.

- "Support Vector Machine" (SVM) is a supervised machine learning algorithm which can be used for both classification or regression challenges. However, it is mostly used in classification problems.
- In this algorithm, we plot each data item as a point in n-dimensional space (where n is number of features you have) with the value of each feature being the value of a particular coordinate.
- Then, we perform classification by finding the hyperplane that differentiate the two classes very well.
- SVM constructs a hyperplane in multidimensional space to separate different classes. SVM generates optimal hyperplane in an iterative manner, which is used to minimize an error.
- The core idea of SVM is to find a maximum marginal hyperplane(MMH) that best divides the dataset into classes.

# Support Vector Machines

**Concept of a Hyperplane in 2D:**

**Hyperplanes**, it is a boundary which 'separates' the data set into its classes (in this case, separates spam emails from the ham ones). It could be lines, 2D planes, or even n-dimensional planes that are beyond our imagination.

A **line** that is used to classify one class from another is also called a **hyperplane**. In fact, it is the model you're trying to build as shown in the figure-1 below-

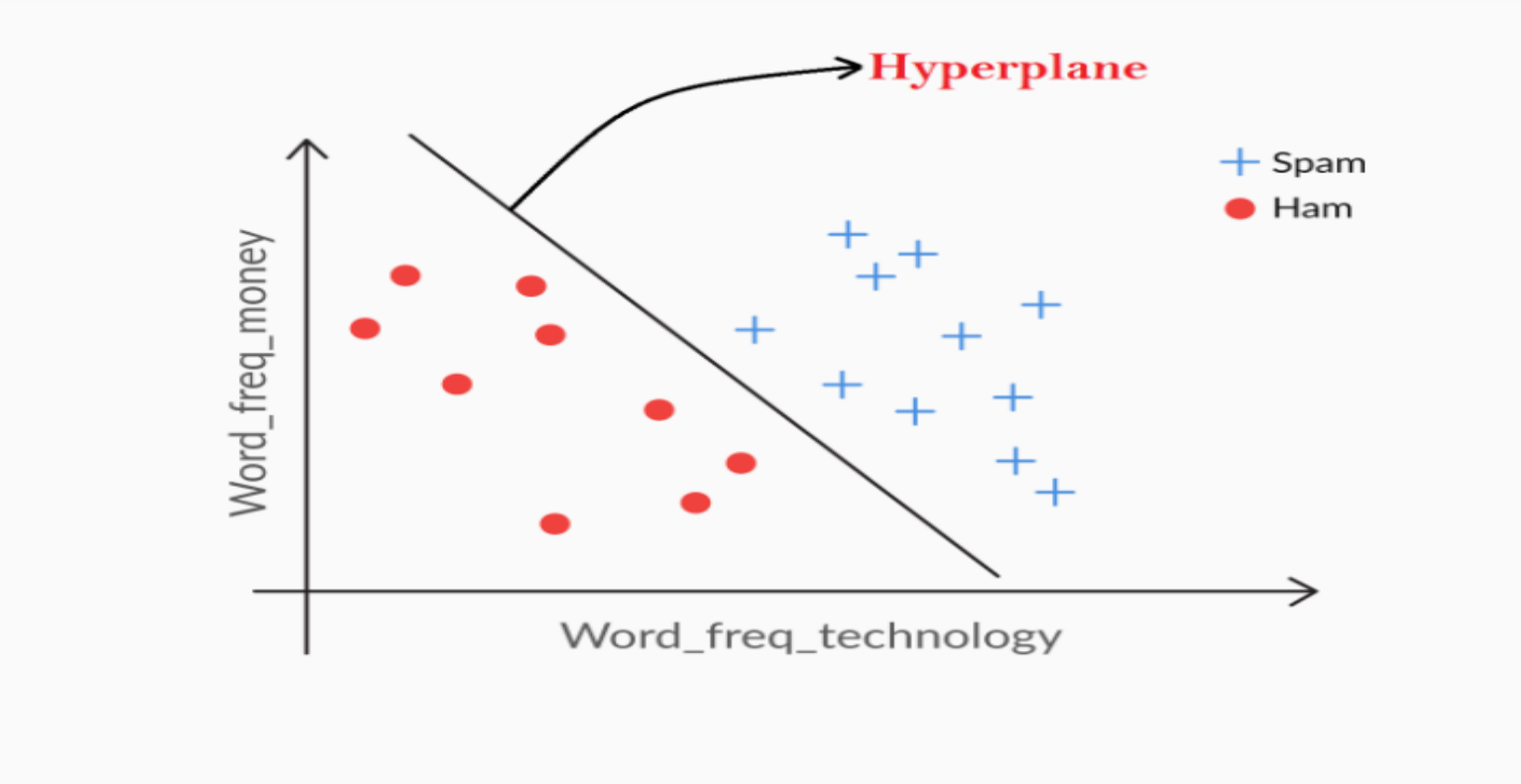
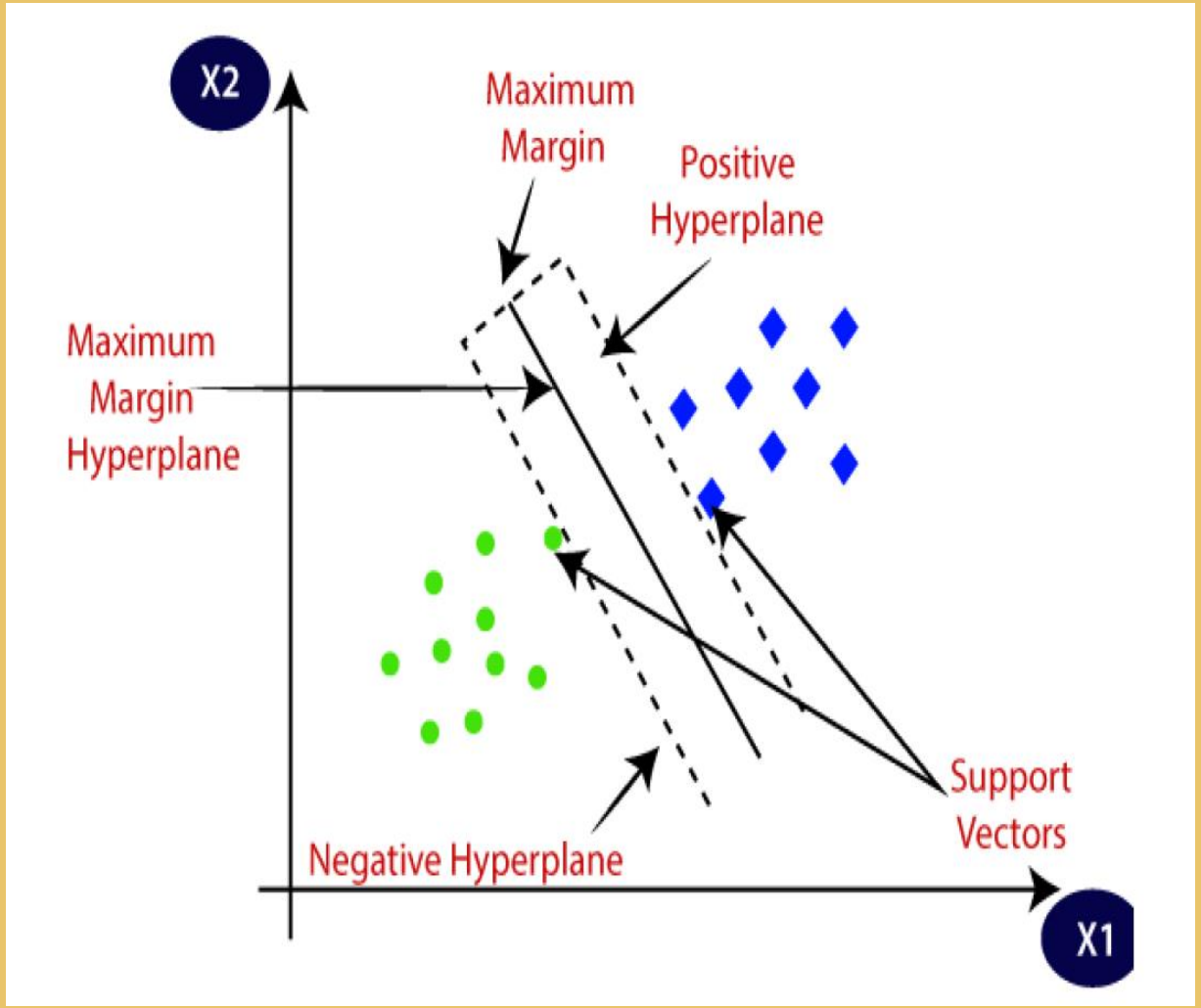



Figure-1: 2D- Hyperplane

# Support Vector Machines

**How Does Support Vector Machine Algorithm Work?**

- The idea behind the SVM classifier is to come up with a hyperplane in an N-dimensional space that divides the data points belonging to different classes.
- However, this hyperplane is chosen based on margin as the hyperplane providing the maximum margin between the two classes is considered.
- These margins are calculated using data points known as Support Vectors.
- Support Vectors are those data points that are near to the hyper-plane and help in orienting it. Support vectors are those data points whose basis the margins are calculated and maximized.



# Maximal Margin Classifier

**There could be multiple lines (Hyperplanes) possible which perfectly separate the two classes as shown in the figure below. But the best line, is the one which maintains the largest possible equal distance from the nearest points of both the classes so for the separator to be optimal, the margin or the distance of the nearest point to the separator should be maximum. This is called Maximal Margin classifier.**

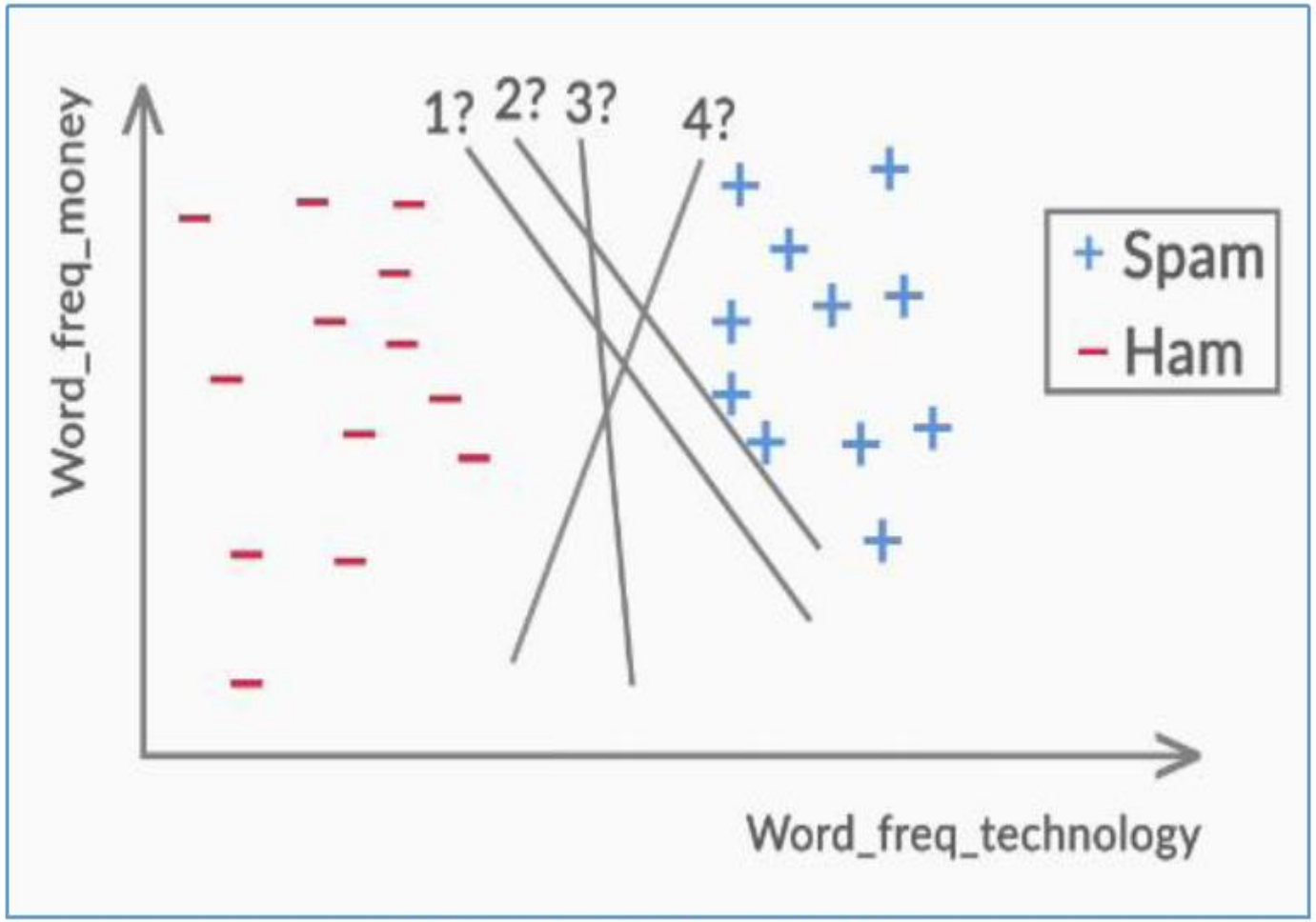
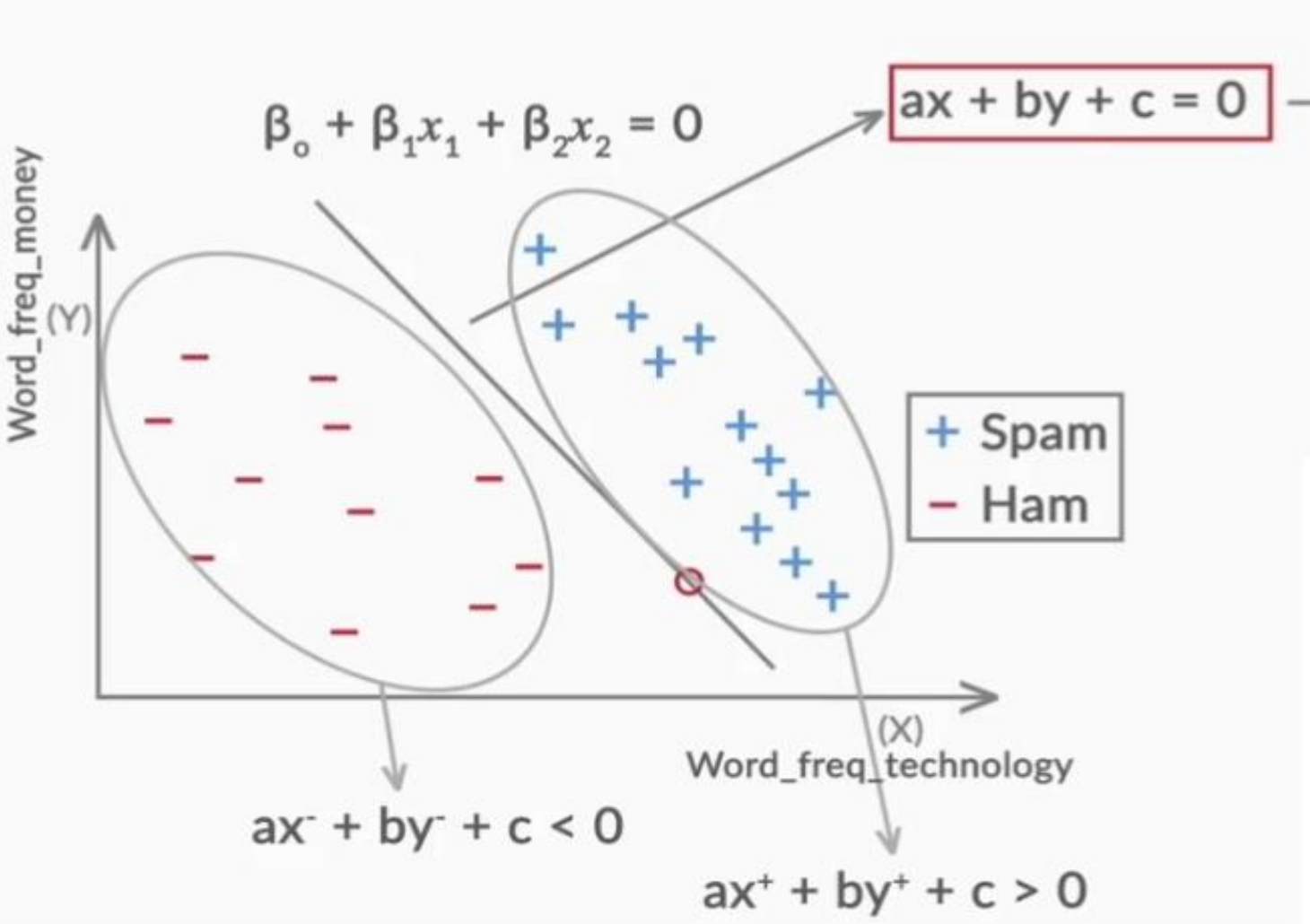



*Figure 5: Maximal Margin Classifier*

# Support Vector Machines

## CHARACTERISATION OF LINE




$ax + by + c = 0 \longrightarrow (a,b,c)$

| X | Y | Z | P | Q | Output |
|---|---|---|---|---|---|
| 3 | 4 | 3 | 8 | 3 | 1 |
| 8 | 9 | 8 | 1 | 2 | 0 |
| 1 | 8 | 6 | 2 | 1 | 0 |
| 3 | 6 | 5 | 5 | 1 | 1 |
| 4 | 1 | 3 | 4 | 0 | 1 |

$ax^- + by^- + c < 0$ Not spam/Ham

$ax^+ + by^+ + c > 0$ Spam

# Support Vector Machines

- The standard equation of a line is given by **ax+by+c = 0**.
- You could generalise it as , $W_0 + W_1 x_1 + W_2 x_2 = 0.$
  - where x1 and x2 are the features — such as 'word_freq_technology' and 'word_freq_money' — and W1 and W2 are the coefficients. For any line with W coefficients, substituting the value of features x1 and x2 in the equation of the line determined by its W coefficients, will return a value.

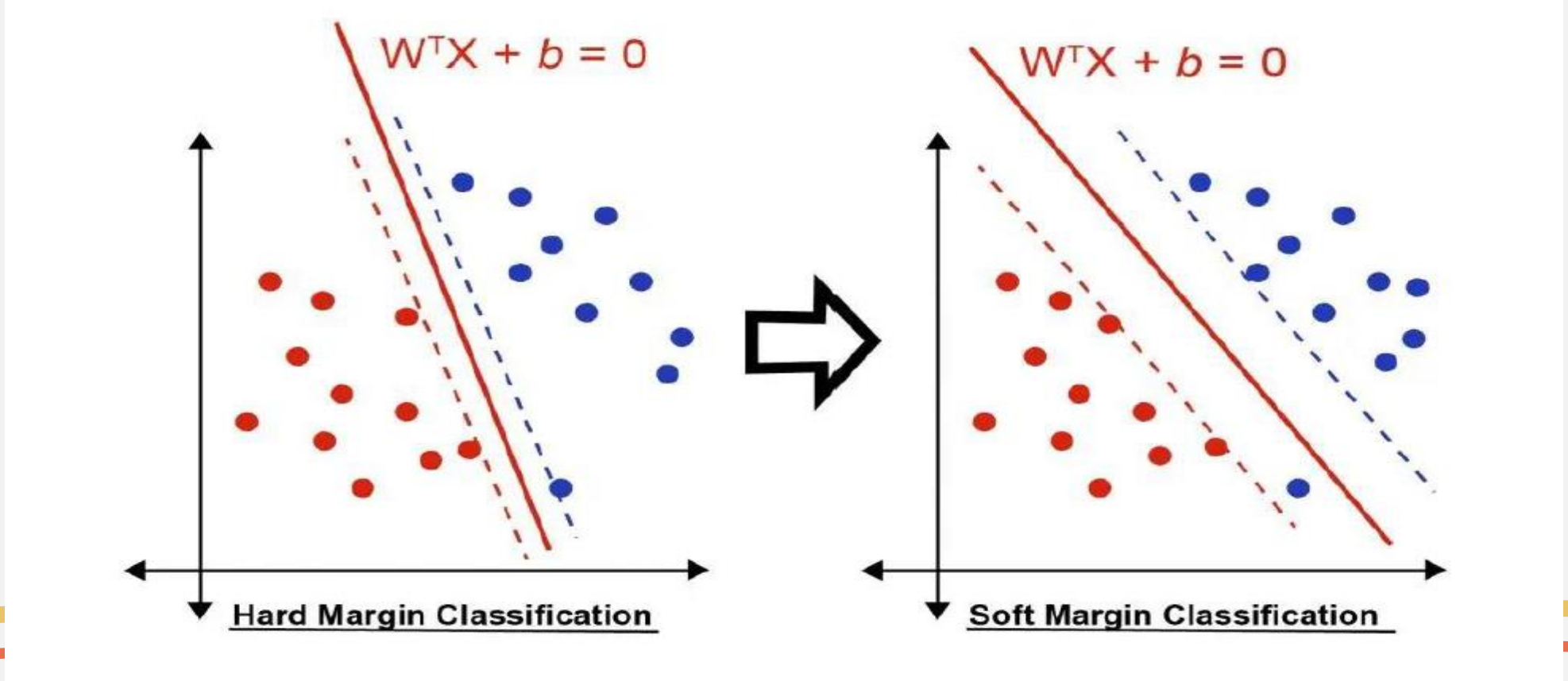
- A **positive value** (blue points in the plot above) would mean that the set of values of the features is in one class; however, a **negative value** (red points in the plot above) would imply it belongs to the other class. A value of zero would imply that the point lies on the line (hyperplane) because any point on the line will satisfy the equation:

$$W_0 + W_1 x_1 + W_2 x_2 = 0.$$

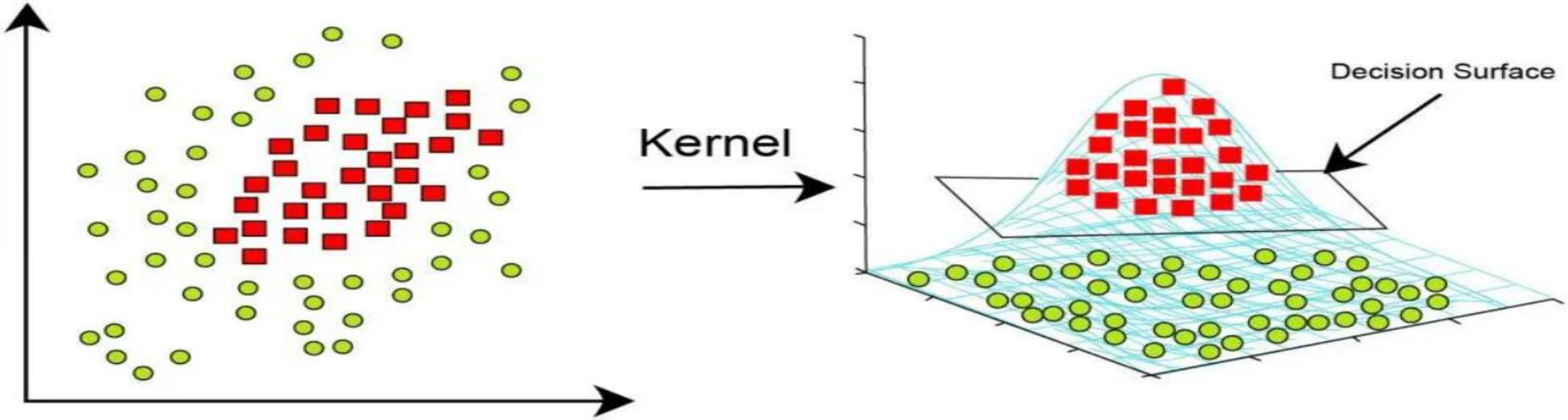
# Support Vector Machines

**Hard Margin:** Hard Margin refers to that kind of decision boundary that makes sure that all the data points are classified correctly. While this leads to the SVM classifier not causing any error, it can also cause the margins to shrink thus making the whole purpose of running an SVM algorithm futile.

**Soft Margin:** Regularization parameter is also added to the loss function in the SVM classification algorithm. This combination of the loss function with the regularization parameter allows the user to maximize the margins at the cost of misclassification. However, this classification needs to be kept in check, which gives birth to another hyper-parameter that needs to be tuned.




Example of Linear SVM through Notes

# Non-Linear SVM and Kernal Function

- The SVM algorithm is implemented in practice using a kernel. A kernel transforms an input data space into the required form.
- SVM uses a technique called the kernel trick. Here, the kernel takes a low-dimensional input space and transforms it into a higher dimensional space.
- In other words, you can say that it converts non-separable problem to separable problems by adding more dimension to it.
- It is most useful in non-linear separation problem. Kernel trick helps you to build a more accurate classifier.